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8. Rate of diffusion $\alpha\ 1/\sqrt{m}$

$m_{NH_3} = 17$; $m_{HCl} = 36.5$

$\gamma_{NH_3} > \gamma_{HCl}$

Hence white fumes first formed near hydrogen chloride

12. Higher the value of 'a', greater the intermolecular force of attraction, easier the liquefaction.

option (c) is correct

14.

$$\text{Compressibility factor (z)} = \frac{PV}{nRT}$$

$$V = \frac{z \times nRT}{P}$$

$$= \frac{\begin{array}{c}0.8697 \times 1 \times 8.314 \\ \times 10^{-2}\ \text{bar dm}^3\ \text{K}^{-1}\ \text{mol}^{-1} \\ \times 400\ \text{K}\end{array}}{71\ \text{bar}}$$

$$V = 0.41\ \text{dm}^3$$

15. T_1 $T_2 = 2T_1$

V_1 $V_2 = 2V_1$

P_1 $P_2 = ?$

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

$$P_2 = \frac{P_1V_1}{T_1} \times \frac{T_2}{V_2} = \frac{P_1V_1}{T_1} \times \frac{2T_1}{2V_1}$$

$$P_2 = P_1$$

Option (c)

16.

$$\frac{\gamma_{H_2}}{\gamma_{C_nH_{2n+2}}} = \sqrt{\frac{m_{C_nH_{2n+2}}}{m_{H_2}}}$$

$$3\sqrt{3} = \sqrt{\frac{m_{C_nH_{2n+2}}}{2}}$$

Squaring on both sides and rearranging

$27 \times 2 = m_{C_nH_{2n-2}}$

$54 = n(12) + (2n-2)(1)$

$54 = 12n + 2n - 2$

$54 = 14n - 2$

$n = (54+2)/14 = 56/14 = 4$

17.

$$\frac{\gamma_{O_2}}{\gamma_{H_2}} = \sqrt{\frac{m_{H_2}}{m_{O_2}}} = \sqrt{\frac{2}{32}} = \frac{1}{4}$$

$$\gamma_{O_2} = \frac{1}{4}\gamma_{H_2}$$

The fraction of oxygen that escapes in the time required for one half of the hydrogen to escape is 1/8

18.

$$\frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_P \quad [\text{For an ideal gas } PV = nRT]$$

$$= \frac{1}{V}\left(\frac{\partial\left(\frac{nRT}{P}\right)}{\partial T}\right)_P$$

$$= \frac{nR}{PV}\left(\frac{\partial T}{\partial T}\right) = \frac{nR}{nRT} = \frac{1}{T}$$